



LIGHT-02

Class 08 - Science

Section A

1. **Fill in the blanks:** [18]
- (a) Muscles attached to the eye lens relax and the lens becomes _____ when distant objects are to be seen. [1]
 - (b) Cones are sensitive to _____ light. [1]
 - (c) The image formed by a plane mirror is called _____ image. [1]
 - (d) The size of the pupil becomes _____ when you see in dim light. [1]
 - (e) The size of the pupil is controlled by the _____. [1]
 - (f) Night birds have _____ cones than rods in their eyes. [1]
 - (g) Splitting of light into its colours is known as _____ of light. [1]
 - (h) Braille system has _____ dot patterns. [1]
 - (i) Kaleidoscope is based on the concept of _____. [1]
 - (j) Reflection from a smooth and shiny surface is called _____ reflection. [1]
 - (k) To keep our eyes fit, our diet should include vitamin _____ rich eatables. [1]
 - (l) If you touch your _____ ear with right hand in front of a plane mirror it will be seen in the mirror that your right ear is touched with _____. [1]
 - (m) Never ever look _____ at the sun. [1]
 - (n) Reflection from a rough surface is called _____ reflection. [1]
 - (o) A person 1 m in front of a plane mirror seems to be _____ m away from his image. [1]
 - (p) Uneven surfaces show _____ reflection. [1]
 - (q) The impression of an image is formed on _____. [1]
 - (r) Angle of reflection is always _____ to the angle of incidence. [1]

Section B

2. Light _____ for propagation. [1]
- a) Do not require medium
 - b) Require denser medium
 - c) Require medium
 - d) Require rarer medium
3. Braille system has _____ dot pattern. [1]
- a) 60
 - b) 63
 - c) 36
 - d) 25
4. Human eye can see maximum _____ separate pictures in a second. [1]
- a) 24
 - b) 12

c) 16

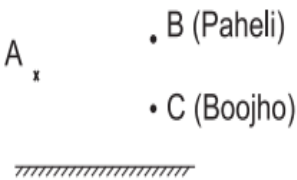
d) 6

Section C

5. Which of the following statements is correct regarding rods and cones in the human eye? [1]
a) Cones are sensitive to bright light b) Rods are sensitive to bright light
c) Cones are sensitive to dim light d) Rods can sense colour
6. Choose the correct option: Angle of incidence is equal to the angle of reflection. [1]
a) Sometimes b) Always
c) Never d) Under special conditions

Section D

7. What is the front transparent parts of the eye called? [1]
8. What is lateral inversion? [1]
9. How many kinds of nerve cells are there in the retina? [1]
10. Give an example of natural dispersion. [1]
11. Who invented the system of reading for blind man? [1]
12. What is the small opening in the iris called? [1]
13. Which cells of retina are sensitive to dim light? [1]
14. What is reflected ray? [1]
15. How many reflected rays can there be for a given single incident ray falling on a plane mirror ? [1]
16. Which cells of the retina are sensitive to bright light and colour? [1]
17. What is incident ray? [1]
18. What is regular reflection? [1]
19. What is the function of iris? [1]
20. How many images of a candle will be formed if it is placed between two parallel plane mirrors separated by 40 cm? [1]
21. What is the coloured part of eye called? [1]
22. Which part of eye is controlled by iris? [1]
23. What is the function of eyelids. [1]
24. What is rectilinear propagation of light? [1]
25. Define angle of incidence. [1]
26. Define regular reflection. [1]
27. Where is the image formed in the eye? [1]
28. Define normal. [1]
29. Define angle of reflection. [1]
30. Define dispersion of light. [1]
31. What do you mean by multiple images? [1]
32. What is blind spot? [1]
33. Define lateral inversion. [1]
34. What is the function of rods and cones in our eye? [1]
35. What do you mean by diffused or irregular reflection? [1]
36. Which part of the body sends the sensation felt to the brain? [1]
37. What is the use of kaleidoscope? [1]

38. What is the distinct vision of normal eye? [1]
39. Name the following: [4]
- The dark muscular part of eyes behind the cornea.
 - The small opening in the iris.
 - The part of eyes which gives it distinctive colour.
 - The nerve which helps in the sense of vision.
40. What do you mean by persistence of vision? How persistence of vision form basis of the principle of cinematography? [4]
41. Boojho while waving his hand very fast in front of his eyes, observes that his fingers appear blurred. What could be the reason for it? [4]
42. What is a periscope? How many mirrors are there in a periscope? Write two uses of periscope. [4]
43. Describe the structure and function of eyes? [4]
44. Describe mirror periscope with diagram. [4]
45. Calculate the number of images formed in two plane mirror inclined at an angle of (i) 90° and (ii) 60° . [4]
46.
 - In how many hours after the death of donor, the eyes should be removed? [4]
 - What kind of persons cannot donate their eyes?
 - Explain why we cannot see our surroundings clearly when we enter a darkened cinema hall from a bright sunshine but our vision improves after some time.
47. What is accommodation? How it is achieved by eye lens and ciliary muscles. [4]
48. What is the power of accommodation of the eye? [4]
49.
 - Find out the position of the image of an object situated at A in the plane mirror (Fig. 13.21). [4]
 - Can Paheli at B see this image?
 - Can Boojho at C see this image?
 - When Paheli moves from B to C, where does the image of A move?
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50. Name the following: [4]
- The layer of tissue that protects light sensitive inner parts of the eye and prevents internal reflection of light
 - Tough opaque sheet of tissues that forms the outer covering of the eye
 - The transparent, watery fluid found in between the cornea and the lens
 - The spot on the retina where an object forms the sharpest image.
51. What makes things visible to us? Why we cannot see a book which is placed: [4]
- behind a wooden screen
 - in a dark room
52. Explain the structure and working of a human eye. [4]
53. Write characteristics of image formed by a plane mirror. [4]
54. Explain all parts of a human eye. [4]
55. Explain the internal structure of human eye. Also discuss the functions of various parts of the eye. [4]
56. Differentiate between regular and diffused reflection. Does irregular reflection mean the failure of the laws of reflection? [4]

